AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT248
Analyzing Data

Mid-term Examination MTA - Makeup
Fall Semester 2025/2026

Date: 4/Dec. /2025

ıber of Exam Pages: 4 ıding this cover sheet)	**Time Allowed:** 1.5 hour

tructions:

- Total Marks: **30**
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part 1: MULTIPLE CHOICE QUESTIONS (6 Marks, 1 for each question)

1. If f(x) is a density function such that $f(x) = ax$ for $0 \leq x \leq 2$ and zero otherwise, then the parameter a is equal to:
   a. 2
   b. 0
   c. 0.5
   d. 1
2. If events H and I in a sample space diagram are independent, what is the probability of the intersection of events H and I?
   a. Joint Probability of event H given event H and I.
   b. Probability of event H given event H.
   c. Probability of event H given event I.
   d. Marginal probability of event H or event I
3. With an average of 4 books sold daily at a local bookstore, what is the probability that exactly 3 books will be sold on a particular day? Assume that *Poisson* distribution applies. [Hint: use $(e = 2.71)$]
   a. 35.95
   b. 575.3
   c. 28.9
   d. 150.2
4. Which one of these 'X' variables is a discrete random variable?
   a. An experiment in chemistry is repeated many times, and X is the time required for a reaction to occur in seconds.
   b. A UPS package is randomly selected, and X is the package's weight in pounds.
   c. d) A student is randomly selected, and X is the distance they must travel in feet to go from their dorm room door to the door of their first class on Monday morning.
   d. A student is randomly selected, and X is the number of correct answers on a two-question multiple-choice quiz.
5. What does the probability density function look like in a uniform distribution?
   a. It is a bell-shaped curve
   b. It is a curve skewed to the right.
   c. It is a curve skewed to the left.
   d. None of the above
6. Assuming a standard normal distribution is appropriate, what is the approximate probability that a z-score is greater than or equal to 2.33? In another way, what is $P(Z^3\ 2.33)$?
   a. 0.0100
   b. 0.1500
   c. 0.0990
   d. 0.0099

Part 2: ESSAY QUESTIONS (25 Marks)

Question 2 (6 Marks):

A random sample of 1170 students who are classified by scholarship type (Full Scholarship, Partial Scholarship, and Self-Paying) and their studying environment (Full time and part time) as given in Table 1.

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Table 1

Sponsorship Type	Full-Time	Part-Time
Full Scholarship	25	70
Partial Scholarship	120	95
Self-Paying	380	480

If a student is picked at random from this group, find the probability if:

(a) The student is a full-time one, given that the student got partial scholarship. **(2 Marks)**

(b) The student does not pay for himself/herself, given that the student is a part-time one. **(2 Marks)**

(c) The student is a full-time one, given that the student got a scholarship (full or partial). **(2 Marks)**

Question 3 (6 Marks):

Consider the number X is a random variable at the start of the sequence, it is impossible to forecast with certainty the number of the trial at which the first success will occur. You can assume that the trials are independent and that the probability of success in each trial is p, where $0 < p < 1$. Note that the random variable X tales the value 1 if the first trial results in a success is given as $P(X = 1) = p$.

a. Write down the probability $P(X = 2)$. **(2 Marks)**

b. Write down the probability $P(X = 3)$. **(2 Marks)**

c. Using the given note and your answers to parts a and b, suggest a general formula for the probability $P(X = x)$. **(2 Marks)**

Question 4 (6 Marks):

Suppose that Nicolaus Bernoulli's model is a good one, and that the gender of children born may be modelled as the outcomes of independent rolls of a 35-sided die with 18 faces marked 'boy' and the other 17 marked 'girl'. According to the model, the distribution of the number of boys in a family of n children is given as:

$$p(x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, 2, \ldots, n.$$

Find the probability that a family of four children:

a. will all be girls; **(3 Marks)**

b. will contain two boys and two girls. **(3 Marks)**

Question 5 (6 Marks):

Use the table given in Appendix 1 (Page 5) to find each of the following probabilities showing a graphical chart of the normal curve with the probability area for each probability.

a. $P(Z \leq 1.00)$ **(2 Marks)**

b. $P(Z \leq 1.96)$ **(2 Marks)**

c. $P(Z \leq 2.25)$ **(2 Marks)**

End of questions

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Appendix A: Areas under the Normal Curve

z	0	1	2	3	4	5	6	7	8	9
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998	0.9998	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.7	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.8	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
4.0	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

MOCK KEY

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